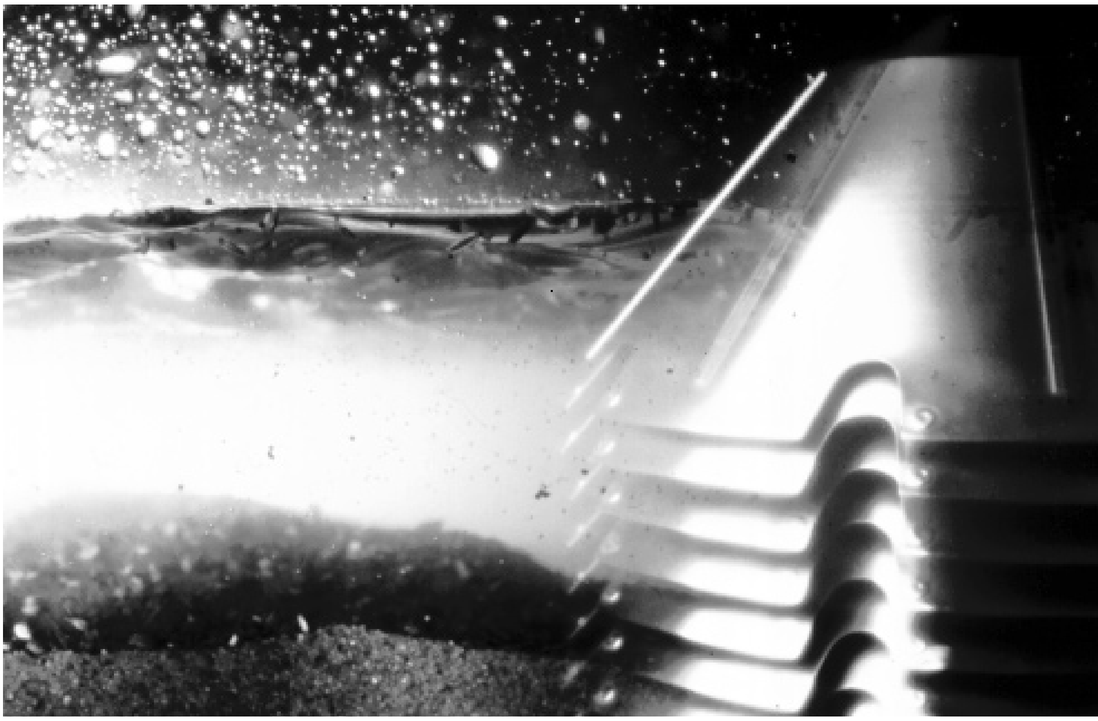
A L F A   L A V A L   S E P A R A T I O N

MAB 104B-14/24C



Separator Manual

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Alfa Laval

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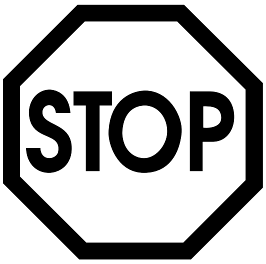

Study instruction manuals and observe the warnings before installation, operation, service and maintenance.

Not following the instructions can result in serious accidents.

In order to make the information clear only foreseeable conditions have been considered. No warnings are given, therefore, for situations arising from the unintended usage of the machine and its tools.

1 Read this first

This manual is designed for operators and service engineers working with the Alfa Laval separator MAB 104B-14/24C.

For information concerning the function of the separator, see ‘‘3 Separator Basics” on page 15, and ‘‘4 Operating Instructions” on page 33.

If the separator has been delivered and installed by Alfa Laval as part of a processing system, this manual is a part of the System Manual. In this case, study carefully all the instructions in the System Manual.

In addition to this Separator Manual a *Spare Parts Catalogue, SPC* is supplied.

This Separator Manual consists of:


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Safety Instructions

Pay special attention to the safety instructions for the separator. Not following the safety instructions can cause accidents resulting in damage to equipment and serious injury to personnel.

Separator Basics

Read this chapter if you are not familiar with this type of separator.

Operating Instructions

This chapter contains operating instructions for the separator only.

Service Instructions

This chapter gives instructions for daily checks, cleaning, oil changes, servicing and check points.

Dismantling / Assembly

This chapter contains step-by-step instructions for dismantling and assembly of the separator for service and repair.

Trouble-tracing

Refer to this chapter if the separator functions abnormally.

If the separator has been installed as part of a processing system always refer to the Trouble-tracing part of the System Manual first.

Technical Reference

This chapter contains technical data concerning the separator and drawings.

Installation

General information on installation planning.

Lifting instruction.

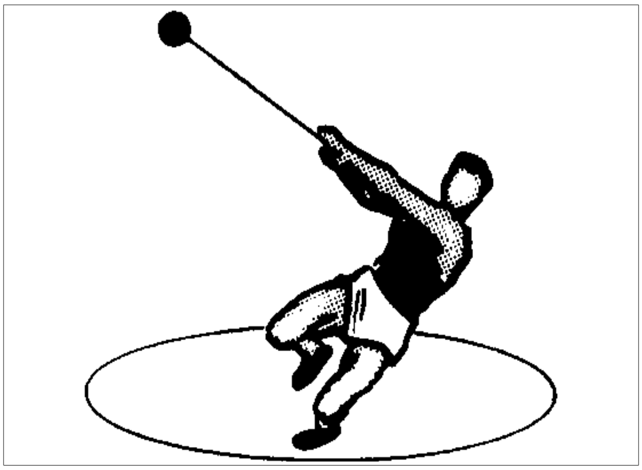
Index

This chapter contains an alphabetical list of subjects, with page references.

2 Safety Instructions

The centrifugal separator includes parts that rotate at high speed. This means that:

- Kinetic energy is high
- Great forces are generated
- Stopping time is long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Manufacturing tolerances are extremely fine. Rotating parts are carefully balanced to reduce undesired vibrations that can cause a breakdown. Material properties have been considered carefully during design to withstand stress and fatigue.

The separator is designed and supplied for a specific separation duty (type of liquid, rotational speed, temperature, density etc.) and must not be used for any other purpose.

Incorrect operation and maintenance can result in unbalance due to build-up of sediment, reduction of material strength, etc., that subsequently could lead to serious damage and/or injury.

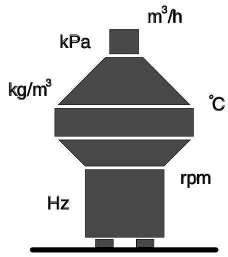
The following basic safety instructions therefore apply:

- **Use the separator only for the purpose and parameter range specified by Alfa Laval.**
- **Strictly follow the instructions for installation, operation and maintenance.**
- **Ensure that personnel are competent and have sufficient knowledge of maintenance and operation, especially concerning emergency stopping procedures.**
- **Use only Alfa Laval genuine spare parts and the special tools supplied.**

DANGER

Disintegration hazards

- Use the separator only for the purpose and parameter range specified by Alfa Laval.
- If excessive vibration occurs, **stop** separator and **keep bowl filled** with liquid during rundown.
- When power cables are connected, always check direction of motor rotation. If incorrect, vital rotating parts could unscrew.
- Check that the gear ratio is correct for power frequency used. If incorrect, subsequent overspeed may result in a serious break down.
- Welding or heating of parts that rotate can seriously affect material strength.
- Wear on the large lock ring thread must not exceed safety limit. ϕ-mark on lock ring must not pass opposite ϕ-mark by more than specified distance.
- Inspect regularly for **corrosion** and **erosion** damage. Inspect frequently if process liquid is corrosive or erosive.



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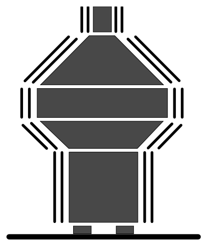
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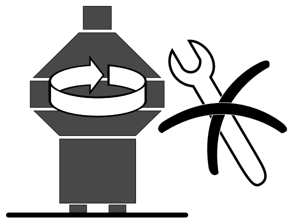
DANGER

Entrapment hazards

- Make sure that rotating parts have come to a **complete standstill** before starting **any** dismantling work.
- To avoid accidental start, switch off and lock power supply before starting **any** dismantling work.
- Assemble the machine **completely** before start. **All** covers and guards must be in place.



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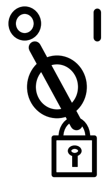

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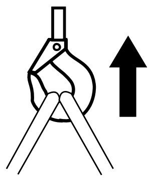
Electrical hazards

- Follow local regulations for electrical installation and earthing (grounding).

WARNING

Crush hazards

- Use correct lifting tools and follow lifting instructions.
- Do **not** work under a hanging load.



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Noise hazards

- Use ear protection in noisy environments.

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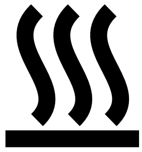
CAUTION

Burn hazards

- Lubrication oil and various machine surfaces can be hot and cause burns.

Cut hazards

- Sharp edges on separator discs and lock ring threads can cause cuts.



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Warning signs in the text

Pay attention to the safety instructions in this manual. Below are definitions of the three grades of warning signs used in the text where there is a risk for injury to personnel.

DANGER

Type of hazard

This type of safety instruction indicates a situation which, if not avoided, could result in **fatal injury** or fatal damage to health.

WARNING

Type of hazard

This type of safety instruction indicates a situation which, if not avoided, could result in **disabling injury** or disabling damage to health.

CAUTION

Type of hazard

This type of safety instruction indicates a situation which, if not avoided, could result in **light injury** or light damage to health.

NOTE

This type of instruction indicates a situation which, if not avoided, could result in damage to the equipment.

3 Separator Basics

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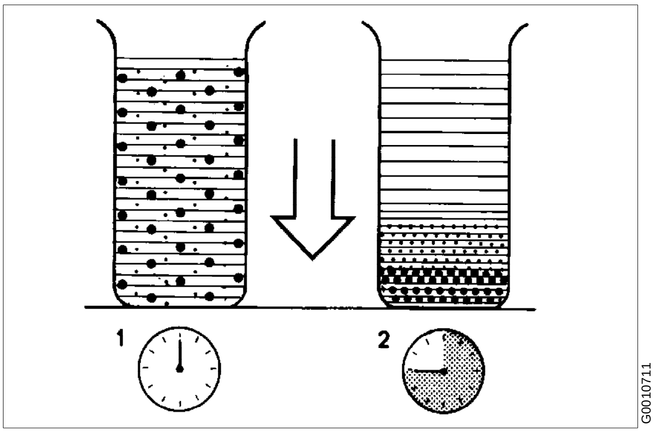
3.1 Basic principles of separation

The purpose of separation can be:

- to free a liquid of solid particles,
- to separate two mutually insoluble liquids with different densities while removing any solids presents at the same time,
- to separate and concentrate solid particles from a liquid.



Sedimentation by gravity

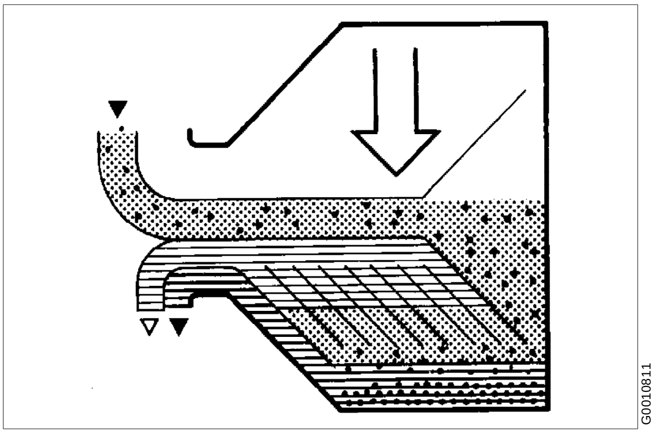
Separation by gravity

A liquid mixture in a stationary bowl will clear slowly as the heavy particles in the liquid mixture sink to the bottom under the influence of gravity.

A lighter liquid rises while a heavier liquid and solids sink.

Continuous separation and sedimentation can be achieved in a settling tank having outlets arranged according to the difference in density of the liquids.

Heavier particles in the liquid mixture will settle and form a sediment layer on the tank bottom.



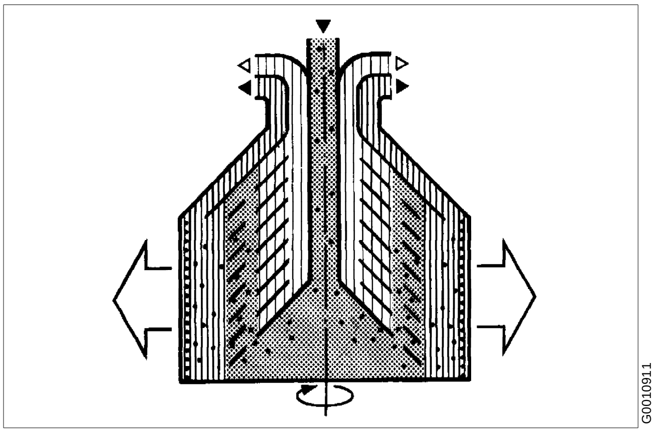
Sedimentation in a settling tank, with outlets making it possible to separate the lighter liquid parts from the heavier

Centrifugal separation

In a rapidly rotating bowl, the force of gravity is replaced by centrifugal force, which can be thousands of times greater.

Separation and sedimentation is continuous and happens very quickly.

The centrifugal force in the separator bowl can achieve in a few seconds what takes many hours in a tank under influence of gravity.

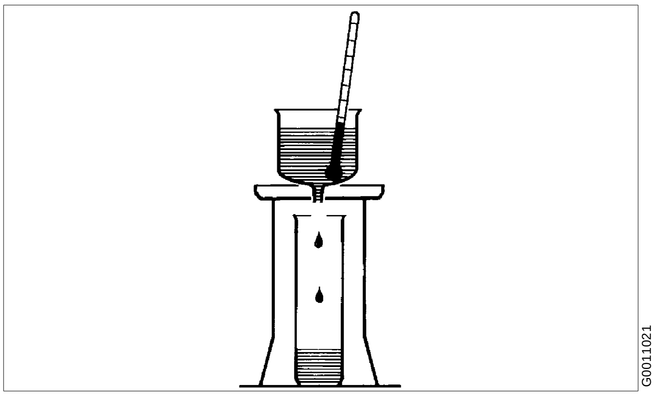


The centrifugal solution

3.1.1 Factors influencing the separation result

Separating temperature

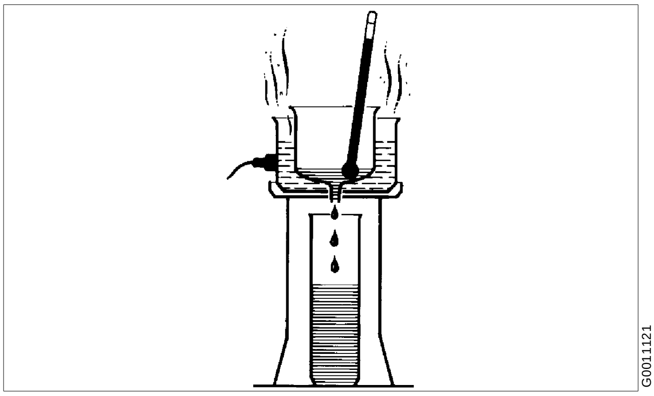
For some types of process liquids (e.g. mineral oils) a high separating temperature will normally increase the separation capacity. The temperature influences oil viscosity and density and should be kept constant throughout the separation.



High viscosity (with low temperature)

Viscosity

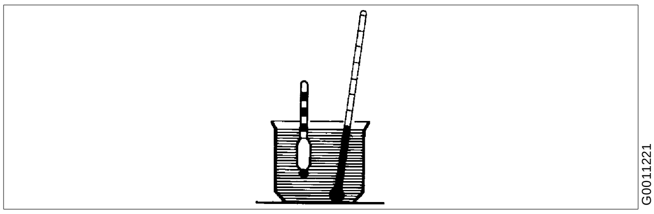
Low viscosity facilitates separation. Viscosity can be reduced by heating.



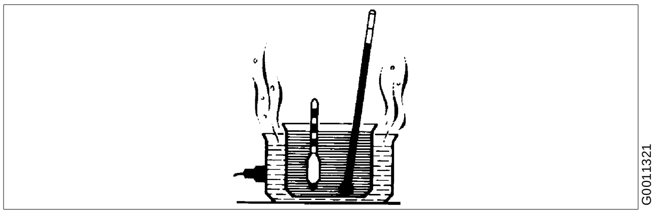
Low viscosity (with high temperature)

Density difference (specific gravity ratio)

The greater the density difference between the two liquids, the easier the separation. The density difference can be increased by heating.



High density (with low temperature)



Low density (with high temperature)

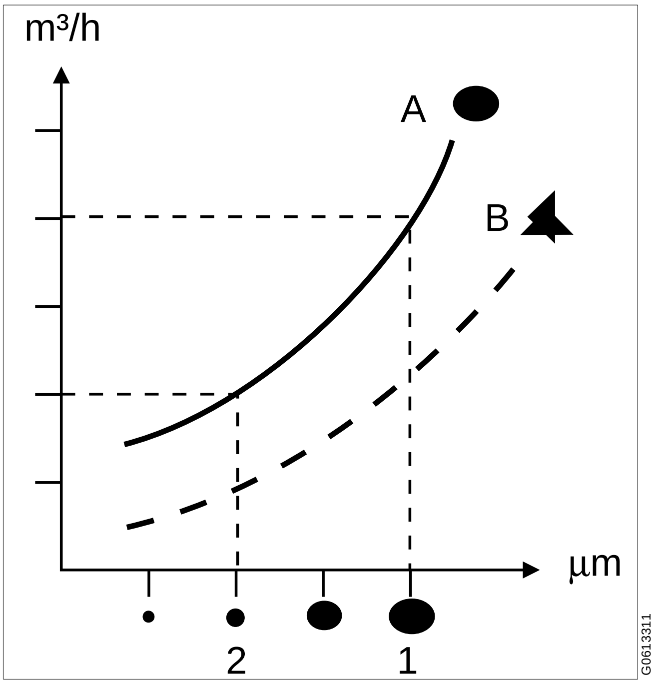
Phase proportions

An increased quantity of water in a oil will influence the separating result through the optimum transporting capacity of the disc stack. An increased water content in the oil can be compensated by reducing the throughput in order to restore the optimum separating efficiency.

Size and shape of particles

The round and smooth particle (A) is more easily separated out than the irregular one (B).

Rough treatment, for instance in pumps, may cause a splitting of the particles resulting in slower separation. Larger particles (1) are more easily separated than smaller ones (2) even if they have the same density.



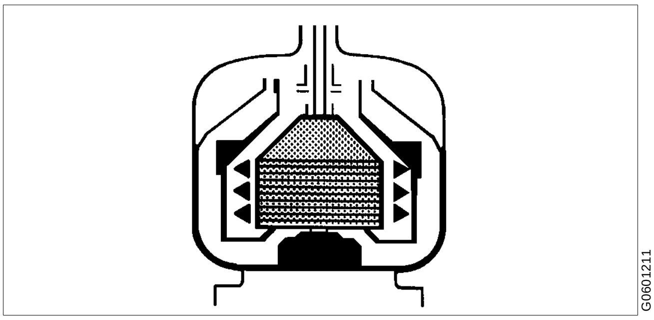
Influence of size and shape

The throughput

The throughput sets the time allowed for the separation of water and sediment from the oil. A better separation result can often be achieved by reducing the throughput, i.e. by increasing the settling time.

Sludge space - sludge content

Sediment will accumulate on the inside periphery of the bowl. When the sludge space is filled up the flow inside the bowl is influenced by the sediment and thereby reducing the separating efficiency. In such cases the time between cleaning should be reduced to suit these conditions.



Sludge accumulation

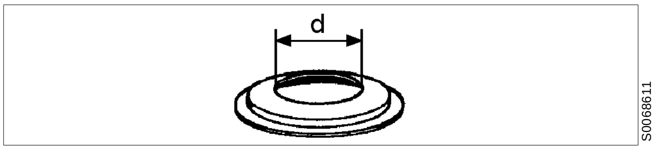
Disc stack

A neglected disc stack containing deformed discs or discs coated with deposits will impair the separating result.

Gravity disc

The position of the interface is adjusted by altering the outlet diameter of the heavy liquid phase, that is by exchanging the gravity disc.

A gravity disc with a larger hole will move the interface towards the bowl perifery, whereas a disc with a smaller hole will place it closer to the bowl centre. For selection of gravity disc see ‘‘3.3.4 Position of interface - gravity disc” on page 24.



Gravity disc diameter adjust the interface position

3.2 Overview

The separator comprises a processing part and a driving part. It is driven by an electric motor.

Mechanically, the separator machine frame is composed of a bottom part and a collecting cover. The motor is flanged to the frame. The frame feet have vibration damping.

The bottom part of the separator contains the horizontal driving device, driving shaft with coupling, a worm gear and a vertical spindle.

The bottom part also contains an oil bath for the worm gear, and a revolution counter, indicating speed.

The collecting cover contain the processing parts of the separator, the inlet and outlets and piping.

The liquid is cleaned in the separator bowl. This is fitted on the upper part of the vertical spindle and rotates at high speed inside the space formed by the collecting cover.

All connections have standardised numbers. These numbers are used in the connection list and the basic size drawing which can be found in chapter ‘‘8 Technical Reference” on page 125.

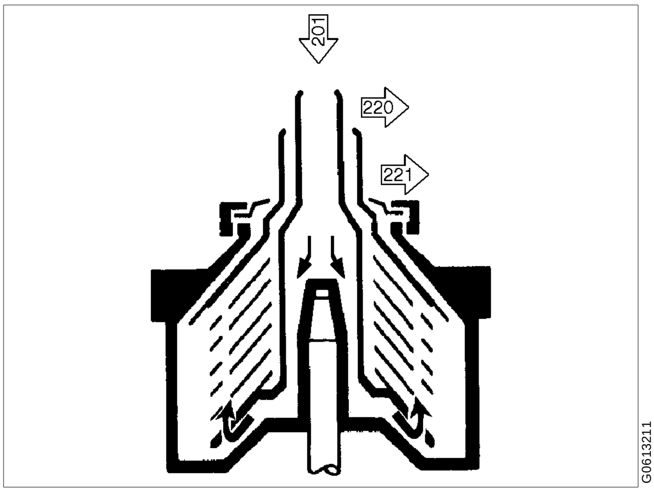
3.3 Separating function

Unseparated process liquid is fed into the bowl through the inlet pipe and is pumped via the distributor towards the periphery of the bowl.

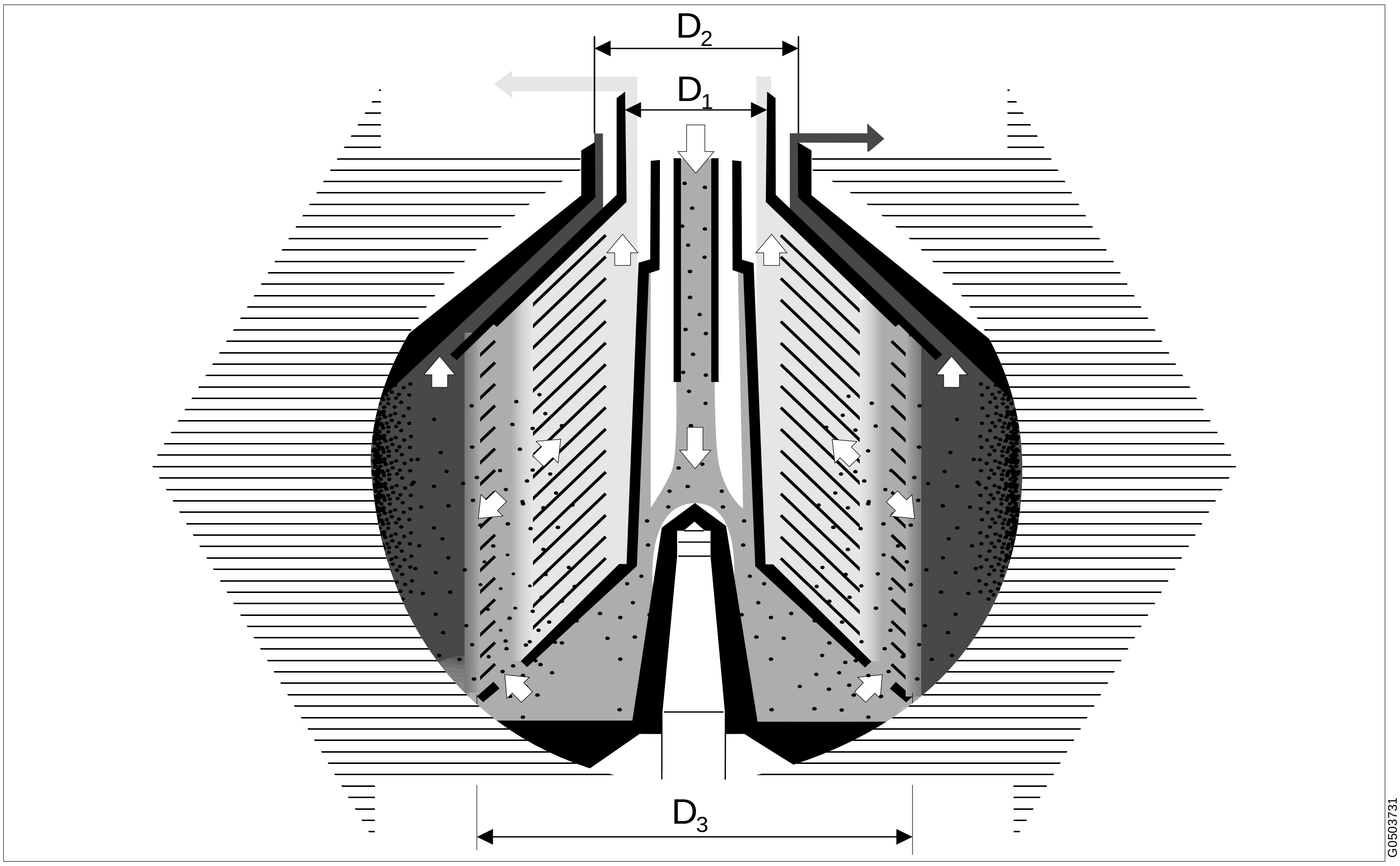
When the process liquid reaches holes of the distributor, it will rise through the channels formed by the disc stack where it is evenly distributed.

The process liquid is continuously cleaned as it flows towards the center of the bowl. When the cleaned process liquid leaves the disc stack it rises upwards and leaves the bowl through outlet (220). Separated heavy phase flows over the gravity disc and leaves the bowl via outlet (221). The sludge and solid particles are forced out towards the periphery of the bowl and collected on the bowl wall, the sludge space.

The space between bowl hood and top disc are normally filled with heavy phase.



3.3.1 Purification



Centrifugal force	Bowl parts
Process liquid	D_1 Diameter of inner outlet.
Heavy liquid phase	D_2 Hole diameter of gravity disc.
Light liquid phase	D_3 Diameter of interface.
Sediment (solids)	

This bowl has two liquid outlets. The process liquid flows through the centre and out under the distributor.

The liquid flows up and is divided among the interspaces between the bowl discs, where the liquid phases are separated from each other by action of the centrifugal force.

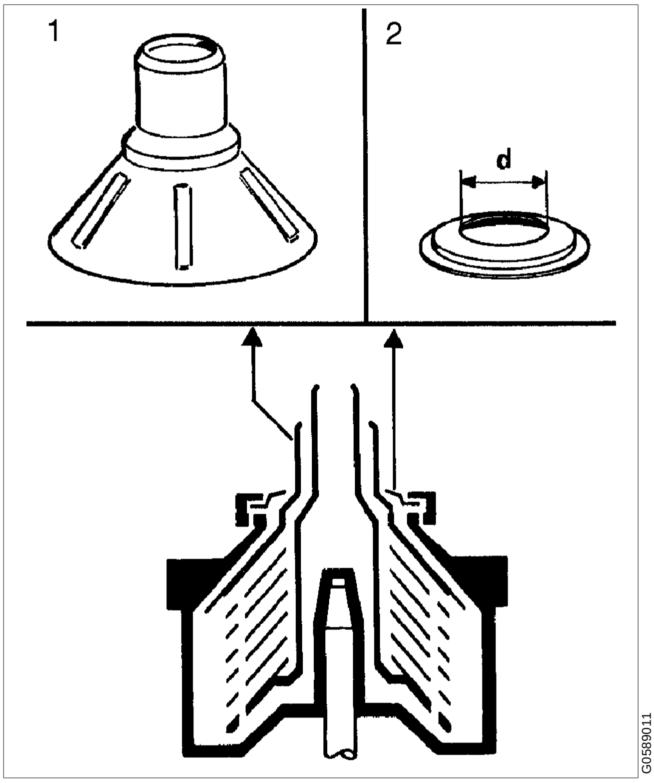
The heavy phase and any sediment move along the underside of the bowl discs towards the perifery of the bowl, where the sediment settle on the bowl wall. The heavy phase proceeds along the upper side of the top disc towards the neck of the bowl hood and leaves the bowl via the gravity disc - *the outer way* (dark coloured in illustration).

The light phase moves along the upper side of the bowl discs towards the bowl centre and leaves the bowl via the hole in the top disc neck - *the inner way* (light coloured in illustration).

3.3.2 Purifier bowl

The figure shows the characteristic parts of a purifier bowl:

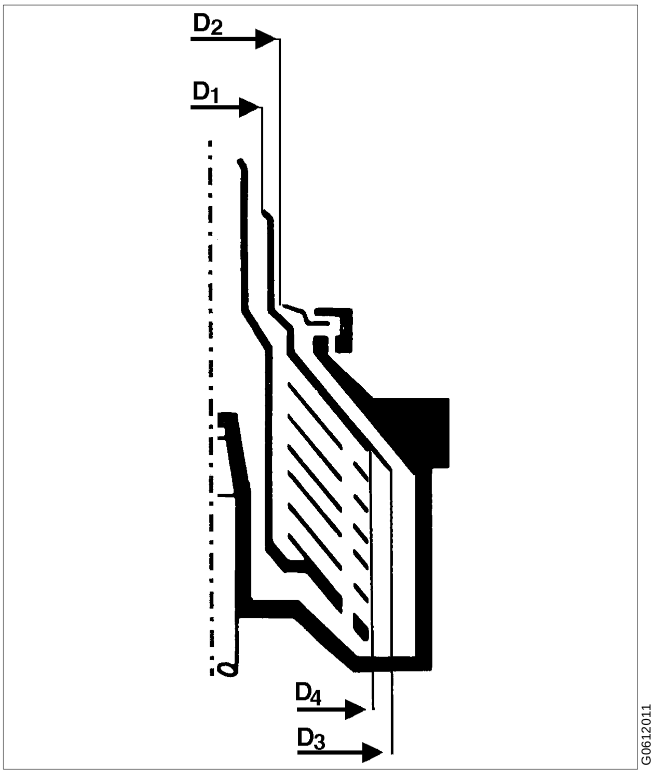
1. Top disc with neck.
2. The gravity disc, which should be chosen according to directions in chapter ''4.1.2 Selection of gravity disc" on page 35.



Purifier bowl

3.3.3 Liquid seal

In a purifier bowl the liquid seal prevents the light liquid phase from passing the outer edge of the top disc, thus eliminating flow out through the outer path. The bowl must therefore be filled with sealing liquid before the process liquid is admitted. The sealing liquid will be displaced slightly by the process liquid into a position that forms the interface. The location of the interface will be affected by the relative difference in density between the phases, but is also dependent on outer and inner diameters (D_1 and D_2 respectively).



The sealing liquid:

- must be insoluble in the light phase.
- must not have higher density than the heavy phase.
- can be soluble in the heavy phase.

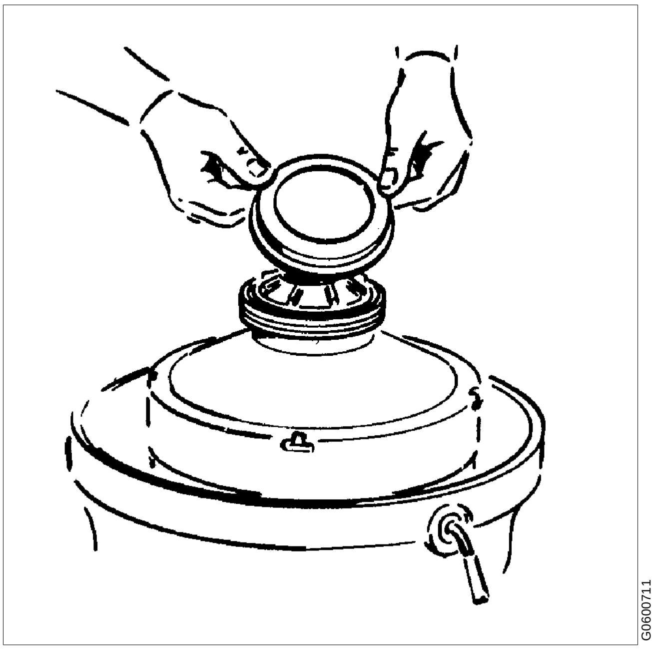
In most cases the heavy phase is used as sealing liquid.

In some cases and only if the process liquid contains a sufficient quantity of heavy phase (more than 25%), the process liquid can be supplied directly as the seal will form automatically.

3.3.4 Position of interface - gravity disc

The position of the interface is adjusted for optimum separation by altering the pressure balance of the two liquid phases oil and water inside the separator.

The purifier bowl is adjusted for separation liquid mixtures with various specific gravity ratios by altering the diameter of the outlet for the heavy phase (D2). The heavier or more viscous the light phase and the larger the liquid feed the smaller the diameter should be. For this purpose a number of gravity discs with various hole diameters is delivered with the separator.



Exchange of gravity disc

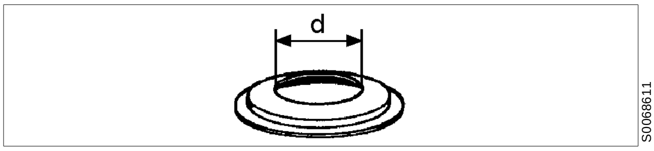
The gravity disc is located inside the bowl hood. A gravity disc with a larger hole will move the interface towards the bowl periphery, whereas a disc with a smaller hole will place it closer to the bowl centre.

In a purifier bowl the position of the interface should be located between the disc stack edge and the outer edge of the top disc.

When selecting a gravity disc for a purifier the general rule is to use the disc having the largest possible hole without causing a break of the water seal.

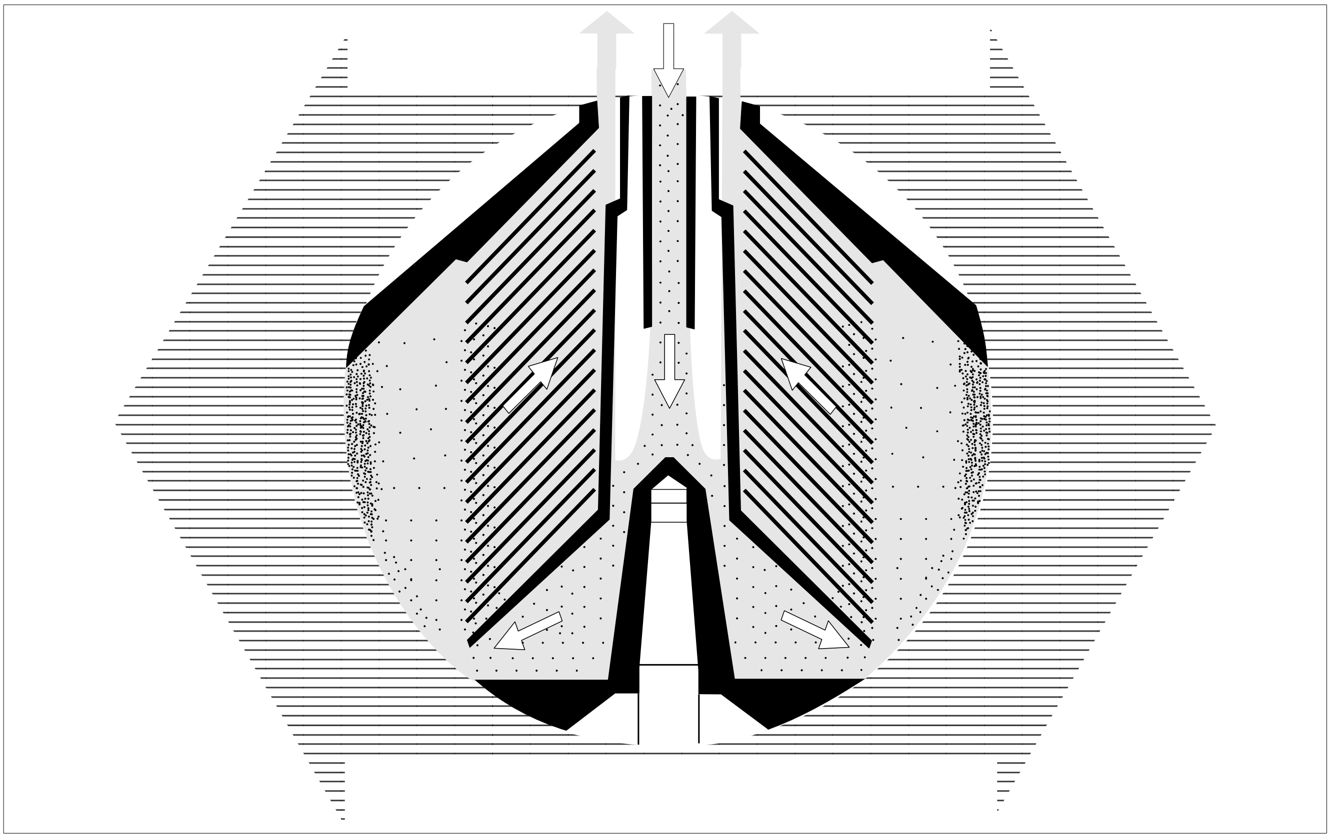
Where to locate the interface depends on which phase should be delivered pure, and on the proportions between the amounts of the two phases as well.

- If the light phase is wanted more free from the heavy one, the interface should be placed nearer the bowl perifery, however not more than the outer edge of the top disc (the gravity disc is too big), as this would break the liquid seal.
- The heavier or more viscous the light phase and the larger the liquid feed the smaller the diameter should be.
- When the heavy phase (water) is wanted more free from the light one (oil), the interface should be placed nearer the bowl centre, however not inside the outer edge of the discs (the gravity disc is too small), as this would prevent the liquid flow.



Gravity disc

3.3.5 Clarification



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Centrifugal force

Process liquid

Bowl parts

Sediment (solids)

This bowl has one liquid outlet.

The process liquid flows through the centre of the distributor.

The liquid flows up and is divided among the interspaces between the bowl discs, where the sediment is separated from the liquid by action of the centrifugal force.

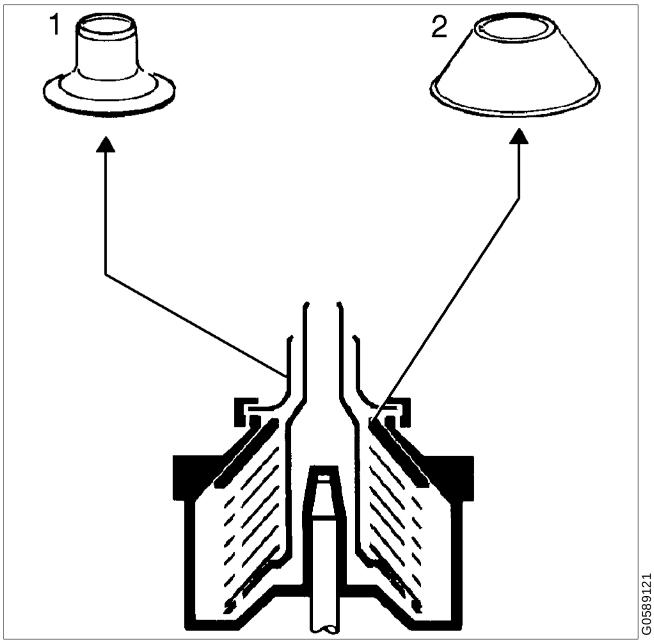
The sediment move along the underside of the bowl discs towards the perifery of the bowl, where it settle on the bowl wall.

The separation is influenced by changes in the viscosity (separating temperature) or in the through-put.

Clarifier bowl

The illustration shows characteristic parts of the clarifier bowl:

1. Discharge collar
2. Bowl disc without caulks

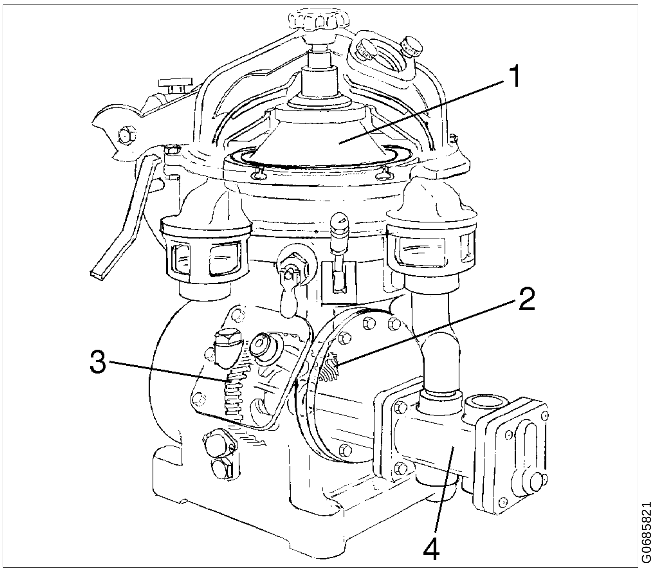


Clarifier bowl

3.4 Mechanical function

3.4.1 Main parts

1. Separator bowl.
2. Vertical driving device.
3. Horizontal driving device.
4. Double pump.

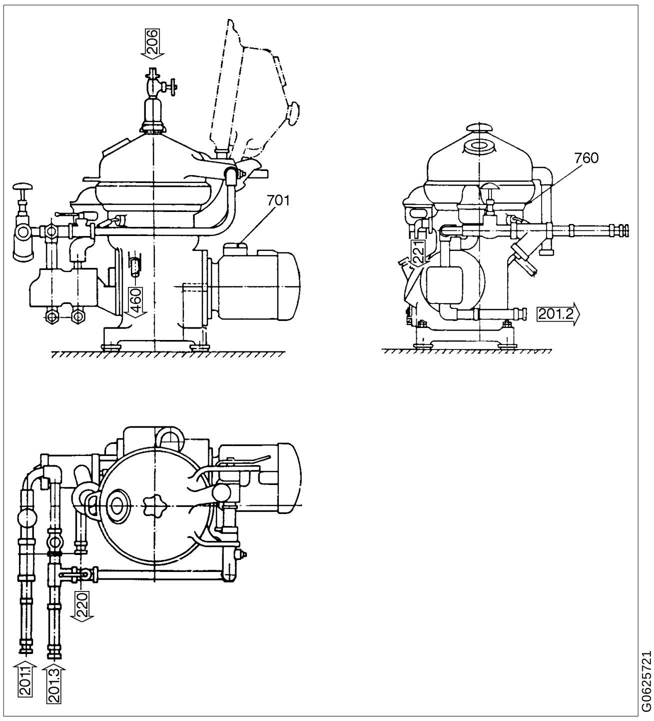


Separator with double pump

3.4.2 Inlet and outlet

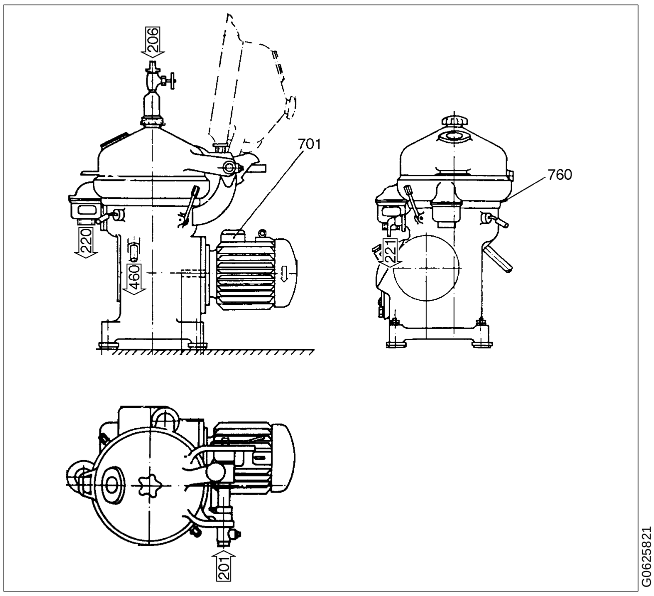
The inlet and outlets consists of the following parts see page 130 to 134.

- The inlet to pump (201.1).
- The outlet (from pump to heater) 201.2.
- The inlet from (heater to separator) 201.3.
- The inlet for water seal (206).
- The outlet for clean oil (220).
- The outlet for water (221).
- Drain of frame (460).



Separator with in- and outlet pump

- The inlet 201.
- The inlet for water seal (206).
- The outlet for clean oil (220).
- The outlet for water (221).
- Drain of frame (460).



Separator without pump

3.4.3 Mechanical power transmission

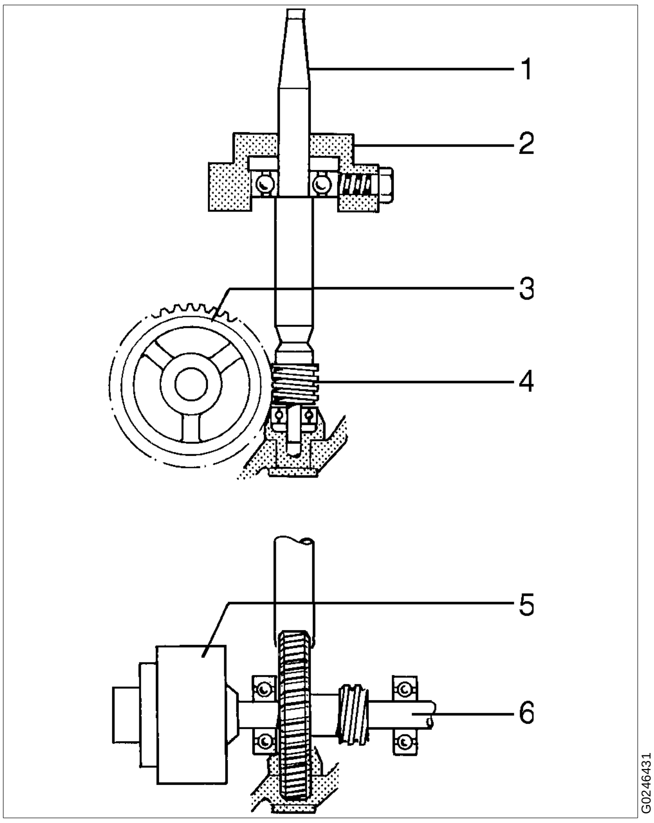
The main parts of the power transmission between motor and bowl are illustrated in the figure.

The flexible coupling allows a certain misalignment between motor and worm wheel shaft.

The worm gear has a ratio which increase the bowl speed several times compared with the motor speed. For correct ratio see chapter ‘‘8.1 Technical data” on page 126.

To reduce bearing wear and the transmission of bowl vibrations to the frame and foundation, the top bearing of the bowl spindle is mounted in a spring casing.

The worm wheel runs in a lubricating oil bath. The bearings on the spindle and the worm wheel shaft are lubricated by the oil splash produced by the rotating worm wheel.

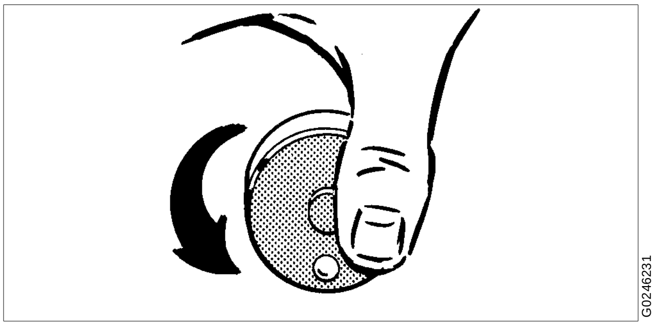


1. *Bowl spindle*
2. *Top bearing and spring casing*
3. *Worm wheel*
4. *Worm*
5. *Flexible coupling*
6. *Worm wheel shaft*

3.4.4 Sensors and indicators

Revolution counter (3)

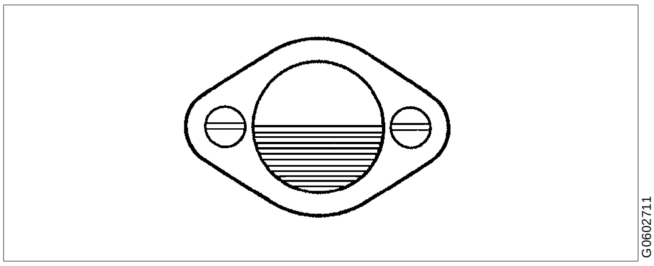
A revolution counter indicates the speed of the separator and is driven from the worm wheel shaft. The correct speed is needed to achieve the best separating results and for reasons of safety. The number of revolutions on the revolution counter for correct speed is shown in chapter "8 Technical Reference" on page 125. Refer to name plate for speed particulars.



Revolution counter - speed indicator

Sight glass (4)

The sight glass shows the oil level in the worm gear housing.



Sight glass - oil level

Cover interlocking switch (6. Option)

When required, the cover interlocking switch should be connected to the starter equipment so that starting of the motor is prevented when the separator hood is not (completely) closed.

3.5 Definitions

Back pressure	Pressure in the separator outlet.
Concentration	Liquid/liquid/solids separation with the intention of separating two intermixed and mutually insoluble liquid phases of different densities. Solids having a higher density than the liquids can be removed at the same time. The heavier liquid phase (water), which is the major part of the mixture, shall be purified as far as possible. The lighter phase must be concentrated.
Clarification	Liquid/solids separation with the intention of separating particles, normally solids, from a liquid having a lower density than the particles.
Clarifier disc	An optional disc, which replaces the gravity disc in the separator bowl, in the case of clarifier operation. The disc seals off the heavy phase (water) outlet in the bowl, thus no liquid seal exists.
Density	Mass per volume unit. Expressed in kg/m^3 at specified temperature, normally at 15 °C.
Gravity disc	Disc in the bowl hood for positioning the interface between the disc stack and the outer edge of the top disc. This disc is only used in purifier mode.
Interface	Boundary layer between the heavy phase (outer) and the light phase (inner) in a separator bowl.
Liquid seal	Liquid in the solids space of the separator bowl to prevent the light phase from leaving the bowl through the heavy phase outlet, in purifier mode.
Major Service (MS)	Overhaul of the complete separator, including bottom part. Renewal of seals in bowl, gaskets in inlet/outlet, seals and bearings in bottom part.
Purification	Liquid/liquid/solids separation with the intention of separating two intermixed and mutually insoluble liquid phases of different densities. Solids having a higher density than the liquids can be removed at the same time. The lighter liquid phase (oil), which is the major part of the mixture, shall be purified as far as possible.
Sediment (sludge)	Solids separated from a liquid.
Throughput	The feed of process liquid to the separator per time unit. Expressed in m^3/h or lit/h.
Viscosity	Fluid resistance against movement. Normally expressed in centistoke (cSt = mm^2/sec), at specified temperature

4 *Operating Instructions*

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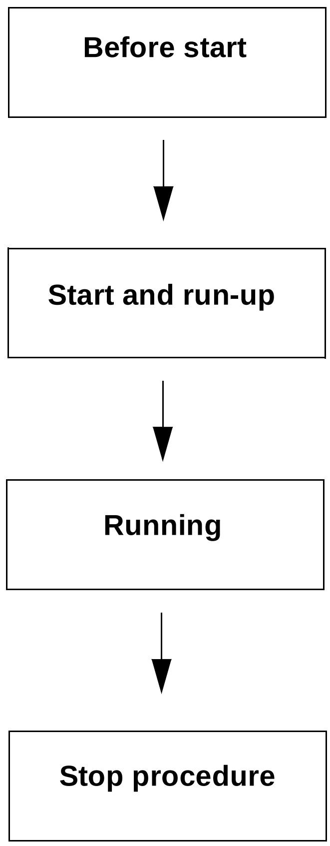
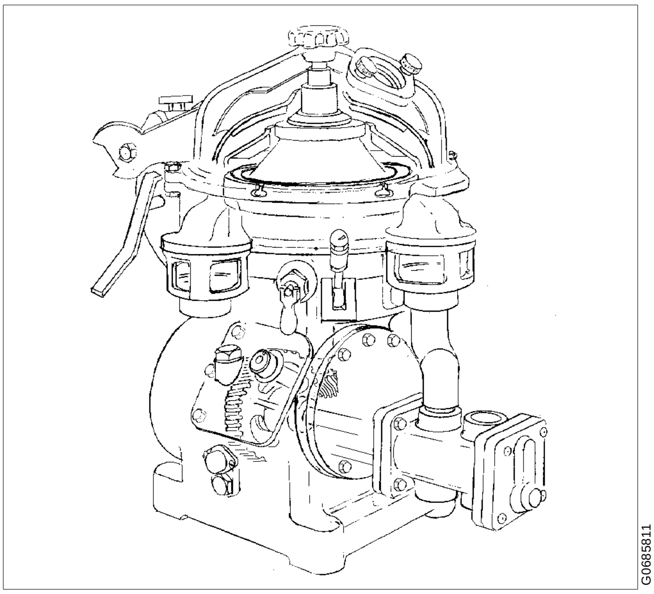
The operating procedure:



4.1 Operating routine

These instructions is related only the separator itself.

> **NOTE**
>
> If there is a system manual, always follow the operating instructions of the system manual. If there is no system manual the instructions below are to be followed



Separator with double pump

4.1.1 Before first start

Technical demands for connections and logical limitations for the separator is described in the chapter ‘‘8 Technical Reference” on page 125 in the documents:

a. Technical data

b. Basic size drawing

c. Connection list

d. Interface description

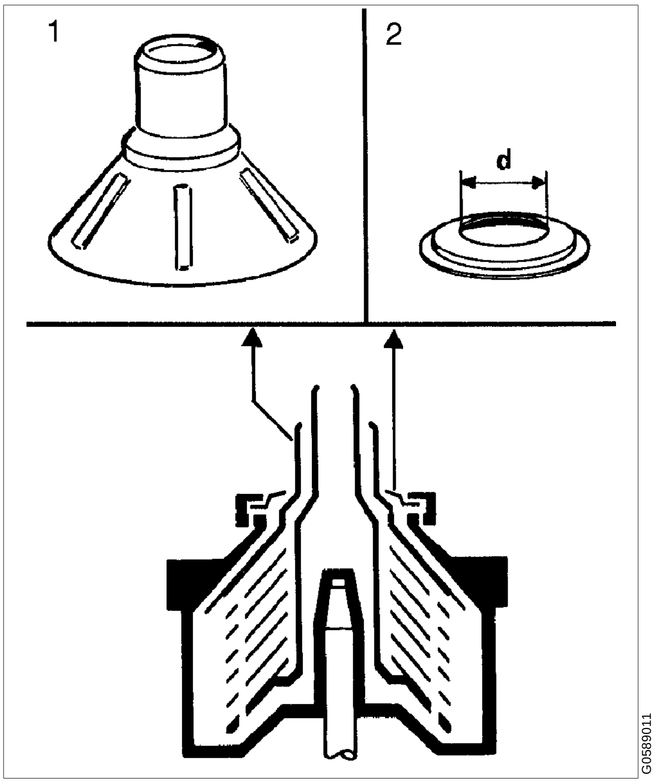
e. Foundation drawing

Before first start the following checkpoint shall be checked:

- Ensure the machine is installed correctly and that feed-lines and drains have been flushed clean.
- Fill oil in the gear housing. Fill up to the middle of the sightglass. Use the correct grade of oil. The separator is delivered without oil in the worm gear housing. For grade and quality, see ‘‘8.8 Lubricants” on page 137

4.1.2 Selection of gravity disc

The separator is delivered with a set of gravity discs. The diameter (d) of the gravity disc (2) sets the position of the interface in the separator. The separation efficiency can be optimized by selection of the correct diameter for each process.

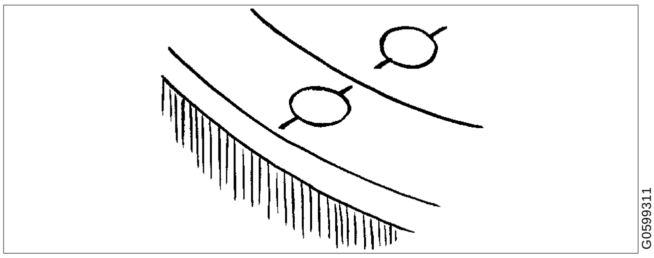


Separator bowl with top disc and gravity disc

4.1.3 Before normal start

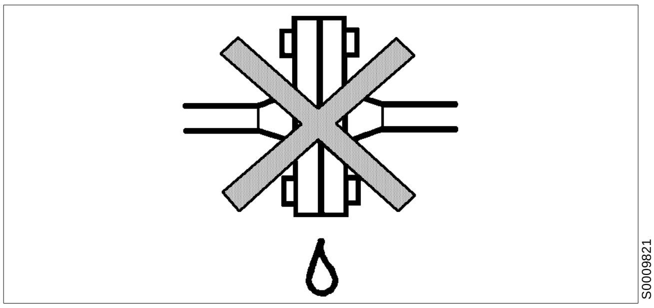
Check these points before every start.

1. Ensure the bowl is clean and that the separator is properly assembled.



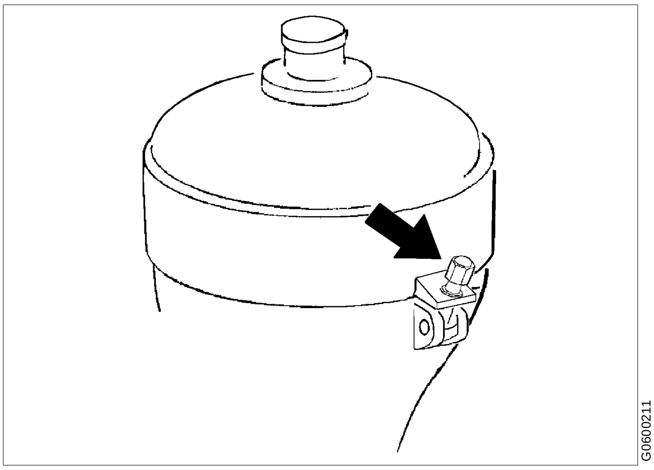
Check separator is assembled

2. Make sure that all inlet and outlet couplings and connections have been correctly made and are properly tightened to prevent leakage.



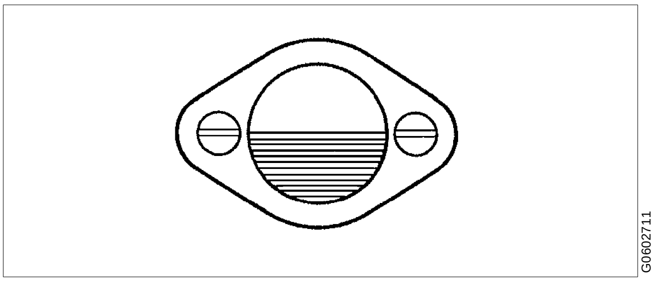
Check pipe connections

3. Check that the hooks and screws for the collecting cover are fully tightened.



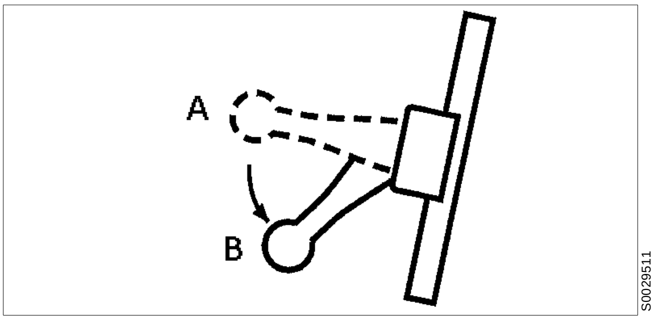
Hook bolts for collecting cover

4. Read the oil level. The middle of the sight glass indicates the **minimum** level. Refill if necessary. For grade and quality, see ‘‘8.8 Lubricants” on page 137.



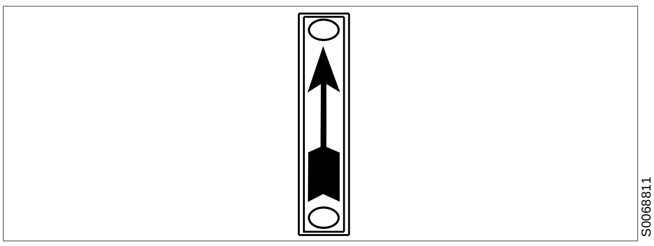
Check oil level

5. Release the brake.



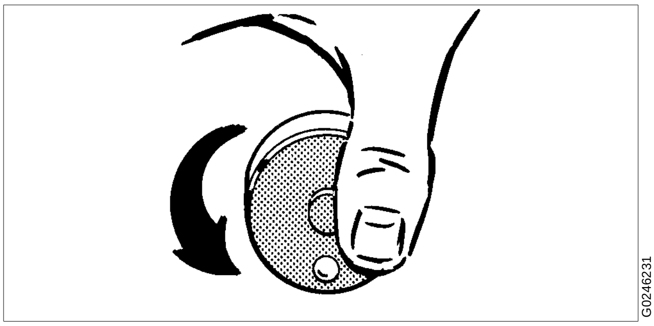
Release the brake

6. Make sure the direction of rotation of the motor and bowl corresponds to the sign on the frame.



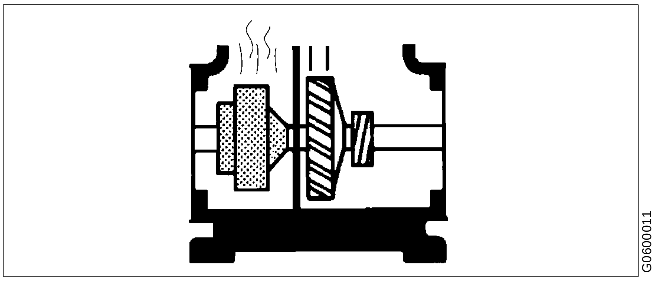
4.1.4 Starting and running-up procedure

1. After starting the separator, visually check to be sure that the motor and separator have started to rotate.
2. Check the direction of rotation. The revolution counter should run anticlockwise.



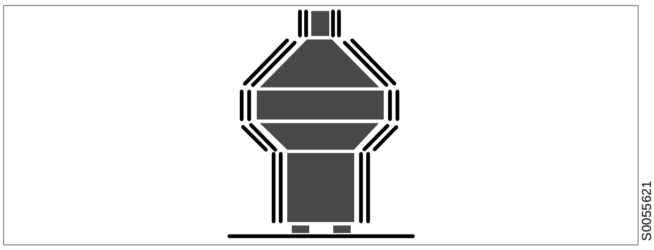
Direction of rotation

3. Be alert for unusual noises or conditions. Smoke and odour may occur at the start when friction pads are new.



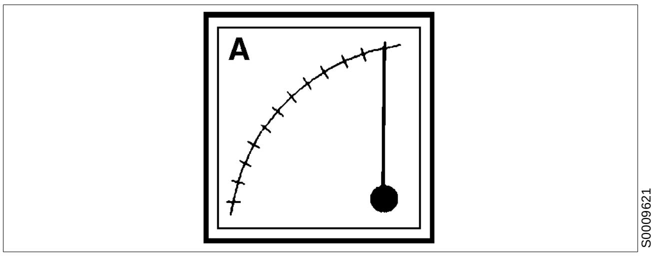
Smoke and odour

4. Note the normal occurrence of critical speed vibration periods.



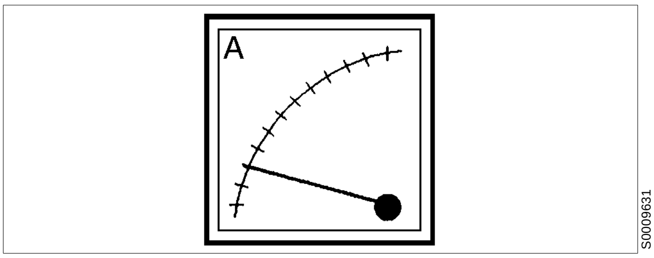
Vibration

5. Motor current indicate when the separator has come to full speed.



Current increases when the coupling engages...

6. During start the current reaches a peak and then slowly drops to a low and stable value. For normal length of the start-up period see ‘‘8.1 Technical data” on page 126.



.... to decrease to a stable value when full speed has been reached.

DANGER

Disintegration hazards

When excessive vibration occurs, **stop** separator and **keep bowl filled** with liquid during rundown.

The cause of the vibrations must be identified and corrected before the separator is restarted.

Excessive vibrations may be due to incorrect assembly or poor cleaning of the bowl.

4.1.5 At full speed

1. Supply liquid (water) to form the liquid seal. Continue until liquid (water) flows out through the heavy phase (water) outlet. The liquid (water) should have the same temperature as the process liquid and be supplied quickly.
2. Close the liquid (water) feed.
3. Start the oil feed slowly to avoid breaking the water seal. Fill the bowl as quickly as possible.
4. Adjust to desired throughput.

4.1.6 During operation

Do regular checks on:

- feed inlet temperature (if applicable)
- collecting tank level
- sound/vibration of the separator
- motor current.

4.1.7 Stopping procedure

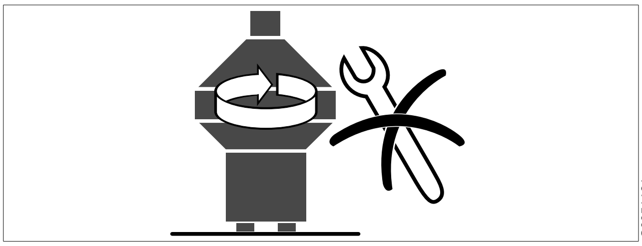
1. Feed sealing water.
2. Turn off the feed.
3. Stop the separator.
4. Manual cleaning should be carried out before next start up. See ‘‘4.2.1 Removal of separated sludge’’.

DANGER

Entrapment hazards

Make sure that rotating parts have come to a **complete standstill** before starting **any** dismantling work.

The revolution counter and the motor fan indicate if the separator is rotating or not.



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